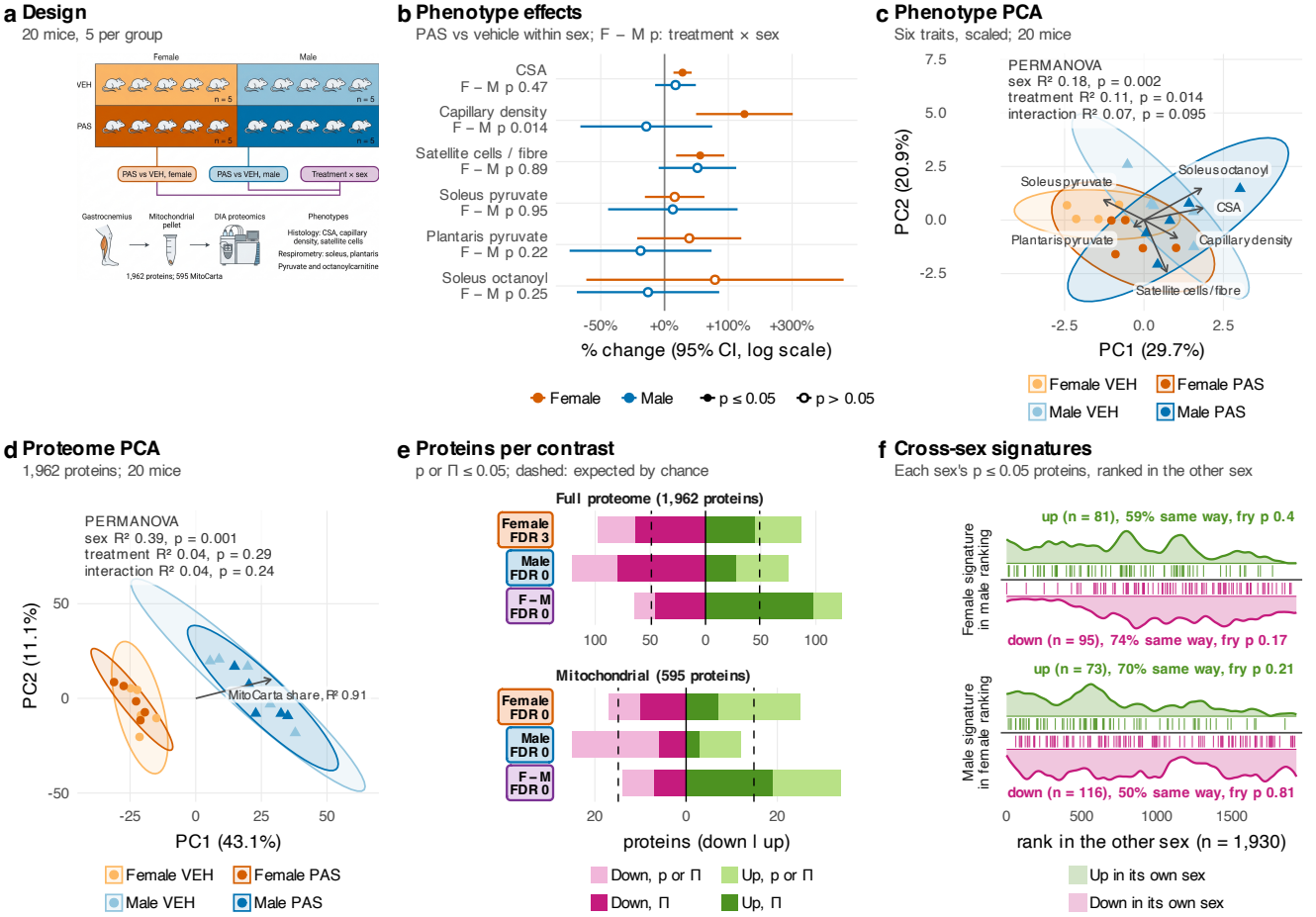

a Design
20 mice, 5 per group
Female
Male
VEH
PAS
n = 5
PAS vs VEH, female
PAS vs VEH, male
Treatment × sex
Gastrocnemius
Mitochondrial pellet
DIA proteomics
Phenotypes
Histology: CSA, capillary density, satellite cells
Respirometry: soleus, plantaris
Pyruvate and octanoylcarnitine
1,962 proteins; 595 MitoCarta
b Phenotype effects
PAS vs vehicle within sex; F – M p: treatment × sex
CSA
F – M p 0.47
Capillary density
F – M p 0.014
Satellite cells / fibre
F – M p 0.89
Soleus pyruvate
F – M p 0.95
Plantaris pyruvate
F – M p 0.22
Soleus octanoyl
F – M p 0.25
-50%
+0%
+100%
+300%
% change (95% CI, log scale)
Female
Male
p ≤ 0.05
p > 0.05
c Phenotype PCA
Six traits, scaled; 20 mice
PERMANOVA
sex R² 0.18, p = 0.002
treatment R² 0.11, p = 0.014
interaction R² 0.07, p = 0.095
Soleus pyruvate
Soleus octanoyl
CSA
Plantaris pyruvate
Capillary density
Satellite cells / fibre
PC2 (20.9%)
PC1 (29.7%)
Female VEH
Female PAS
Male VEH
Male PAS
d Proteome PCA
1,962 proteins; 20 mice
PERMANOVA
sex R² 0.39, p = 0.001
treatment R² 0.04, p = 0.29
interaction R² 0.04, p = 0.24
MitoCarta share, R² 0.91
PC2 (11.1%)
PC1 (43.1%)
Female VEH
Female PAS
Male VEH
Male PAS
e Proteins per contrast
p or Π ≤ 0.05; dashed: expected by chance
Full proteome (1,962 proteins)
Female FDR 3
Male FDR 0
F – M FDR 0
Mitochondrial (595 proteins)
Female FDR 0
Male FDR 0
F – M FDR 0
proteins (down | up)
Down, p or Π
Up, p or Π
Down, Π
Up, Π
f Cross-sex signatures
Each sex's p ≤ 0.05 proteins, ranked in the other sex
up (n = 81), 59% same way, fry p 0.4
down (n = 95), 74% same way, fry p 0.17
Female signature in male ranking
up (n = 73), 70% same way, fry p 0.21
down (n = 116), 50% same way, fry p 0.81
Male signature in female ranking
rank in the other sex (n = 1,930)
Up in its own sex
Down in its own sex